

Foundation (Jalapeno)

- Q1.** A basketball player scores  $x$  points from field goals and  $y$  points from free throws. If the player scores a total of 30 points and each field goal is worth 2 points and each free throw is worth 1 point, which inequality represents the situation?
- (A)  $2x + y \leq 30$   
(B)  $2x + y \geq 30$   
(C)  $2x + y = 30$   
(D)  $x + 2y \leq 30$   
(E)  $x + 2y \geq 30$
- Q2.** A soccer team has 120 minutes to play a game. If they have played  $x$  minutes of offense and  $y$  minutes of defense, which inequality represents the time remaining?
- (A)  $x + y \leq 120$   
(B)  $x + y \geq 120$   
(C)  $x - y \leq 120$   
(D)  $x - y \geq 120$   
(E)  $2x + 2y \leq 120$
- Q3.** A cyclist rides  $x$  miles on flat ground and  $y$  miles uphill. If the cyclist's speed on flat ground is 20 mph and uphill is 10 mph, and the total time is 2 hours, which inequality represents the situation?
- (A)  $\frac{x}{20} + \frac{y}{10} \leq 2$   
(B)  $\frac{x}{20} + \frac{y}{10} \geq 2$   
(C)  $\frac{x}{10} + \frac{y}{20} \leq 2$   
(D)  $\frac{x}{10} + \frac{y}{20} \geq 2$   
(E)  $x + y \leq 40$
- Q4.** A tennis player wins  $x$  games and loses  $y$  games. If the player wins at least 4 more games than loses, which inequality represents the situation?
- (A)  $x - y \geq 4$   
(B)  $x - y \leq 4$   
(C)  $x + y \geq 4$   
(D)  $x + y \leq 4$   
(E)  $2x - 2y \geq 4$
- Q5.** A volleyball team has a budget of 500 for equipment and travel. If the team spends  $x$  dollars on equipment and  $y$  dollars on travel, which inequality represents the budget constraint?
- (A)  $x + y \leq 500$   
(B)  $x + y \geq 500$   
(C)  $x - y \leq 500$   
(D)  $x - y \geq 500$   
(E)  $2x + 2y \leq 500$
- Q6.** A swimmer swims  $x$  laps in the morning and  $y$  laps in the afternoon. If the swimmer swims at most 20 laps per day, which inequality represents the situation?
- (A)  $x + y \leq 20$   
(B)  $x + y \geq 20$   
(C)  $x - y \leq 20$   
(D)  $x - y \geq 20$   
(E)  $2x + 2y \leq 20$
- Q7.** A football player runs  $x$  yards on plays and  $y$  yards on kick-offs. If the player runs at most 100 yards per game, which inequality represents the situation?
- (A)  $x + y \leq 100$   
(B)  $x + y \geq 100$   
(C)  $x - y \leq 100$   
(D)  $x - y \geq 100$   
(E)  $2x + 2y \leq 100$
- Q8.** A gymnast performs  $x$  routines on the floor and  $y$  routines on the beam. If the gymnast performs at most 12 routines per competition, which inequality represents the situation?
- (A)  $x + y \leq 12$   
(B)  $x + y \geq 12$   
(C)  $x - y \leq 12$   
(D)  $x - y \geq 12$   
(E)  $2x + 2y \leq 12$

Open Ended Questions (FRQ)

- Q9.** A baseball player hits  $x$  home runs and  $y$  singles. If the player hits at least 10 more singles than home runs and at most 50 total hits, write a system of inequalities to represent the situation.
- Q10.** A track athlete runs  $x$  miles at a speed of 8 mph and  $y$  miles at a speed of 6 mph. If the athlete runs for at most 2 hours and covers a total distance of at least 10 miles, write a system of inequalities to represent the situation.

Chef's Tips

- Tip 1: Encourage students to graph the inequalities on a coordinate plane.
- Tip 2: Remind students to consider the context of the problem when interpreting the inequalities.
- Tip 3: Suggest that students use different colors to distinguish between different inequalities on the graph.